Haystack Pipeline Architecture

Cynaris AI/ML Internship - Week 7 Reference Material

1. Introduction

Haystack is an open-source framework for building modular AI applications, search systems and retrieval augmented generation (RAG) pipelines.

2. Pipeline Architecture

A Haystack pipeline connects independent components through named inputs and outputs. A component can convert documents, preprocess text, write to a document store, retrieve relevant content or generate answers.

3. Document Conversion

PDF converters extract textual content from PDF files and represent the result as Documents. Metadata such as filename and page number helps preserve source information.

4. Document Stores

A document store manages documents and provides an interface for writing and retrieving them. InMemoryDocumentStore is useful for demonstrations and experiments. Persistent stores are preferable for production workloads.

5. Retrieval-Augmented Generation

A typical RAG workflow converts source documents, indexes them, retrieves relevant passages for a user query and passes the retrieved context to a language model.

6. Pipeline Connections

Pipeline connections must match component input and output types. Components should be tested independently before connecting the complete workflow.

7. Evaluation

A retrieval system should be evaluated using relevant documents, ranking quality, retrieval latency and repeatable test queries. Retrieval quality and generated answer quality are separate evaluation concerns.

8. Best Practices

Keep source metadata, avoid indexing duplicate documents, use consistent preprocessing, record model versions and save reproducible evaluation evidence.